

Titration 2

Essential Pre-Uni Chemistry B7.2

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** GCSE C3, A Level P1

2.50 g of an unknown carbonate were dissolved in 100 cm^3 of 1.00 mol dm^{-3} hydrochloric acid (an excess). The resulting solution was made up to 250 cm^3 in a volumetric flask. 25.00 cm^3 aliquots of this solution were titrated against 0.250 mol dm^{-3} sodium hydroxide. Some of the results are shown below. Fill in the gaps in the table (Parts A-D), and then calculate the quantities in Parts E-L to identify the cation (Part M).

Titration	Initial burette reading / cm^3	Final burette reading / cm^3	Titre / cm^3
Rough	0.60	25.10	Part A
1	0.15	Part B	24.10
2	Part C	25.25	24.45
3	1.35	25.45	Part D

Part A

Rough, Titre/ cm^3

Give your answer to 4 significant figures.

Part B

1, Final burette reading/ cm^3

Give your answer to 4 significant figures.

Part C

2, Initial burette reading/cm³

Give your answer to 2 significant figures.

Part D

3, Titre/cm³

Give your answer to 4 significant figures.

Part E

Average concordant titre

Calculate the average concordant titre. Give your answer to 4 significant figures.

Part F

Amount of sodium hydroxide

Calculate the amount of sodium hydroxide in that volume. Give your answer to 3 significant figures.

Part G

Amount of hydrochloric acid

The amount of hydrochloric acid in each aliquot. Give your answer to 3 significant figures.

Part H

Initial amount of HCl

Calculate the initial amount of hydrochloric acid added to the carbonate. Give your answer to 3 significant figures.

Part I

Final amount of HCl

Calculate the amount of hydrochloric acid remaining after reaction. Give your answer to 3 significant figures.

Part J

Amount of HCl used

Calculate the amount of hydrochloric acid used in reaction with the carbonate. Give your answer to 3 significant figures.

Part K

Amount of carbonate

Calculate the amount of carbonate in 2.50 g. Give your answer to 3 significant figures.

Part L

Molar mass of carbonate

Calculate the molar mass of the carbonate. Give your answer to 3 significant figures.

Note: assume each formula unit of the carbonate contains exactly one carbonate ion.

Part M

Identity of cation

Identify the cation in the carbonate.

Question deck:

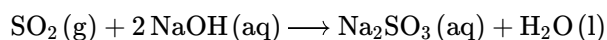
STEM SMART Chemistry Week 16

Titrating Sulfur Dioxide

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level P2

Sulfur dioxide is a by-product of the combustion of coal in power stations. It can react with oxygen and water vapour in the air to form sulfuric acid, H_2SO_4 . This is one of the causes of acid rain.

The amount of sulfur dioxide in the air may be determined by bubbling a sample of the air through sodium hydroxide solution, where it reacts according to the equation below:



The concentration of the unreacted sodium hydroxide can be determined by titration against a standard solution of hydrochloric acid.

1000 dm^3 of air were bubbled through 200 cm^3 of a 1.00 mol dm^{-3} solution of sodium hydroxide. The remaining solution was diluted to 1000 cm^3 with water, and 25.0 cm^3 of this solution was neutralised by 20.4 cm^3 of a 0.100 mol dm^{-3} solution of hydrochloric acid.

Part A

H_2SO_4 formation

Construct an overall equation for the formation of sulfuric acid from sulfur dioxide (do not include state symbols). Balance it so as to use the smallest possible integer coefficients.

Part B

Neutralisation reaction

Give the (net) ionic equation for the reaction of sodium hydroxide with hydrochloric acid.

Part C

Unreacted moles

Find the amount, in moles, of unreacted sodium hydroxide.

Part D

Sulfur dioxide moles

Find the amount, in moles, of sulfur dioxide in 1000 dm^3 of air.

Part E

Percentage by volume

Hence calculate the percentage by volume of sulfur dioxide in air. (You may assume 1 mol of any gas occupies 24 dm^3 at this temperature and pressure.)

Adapted with permission from OCR, A Level Chemistry, June 1999, General and Physical Paper, Question 3

Question deck:

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Titration 4

Essential Pre-Uni Chemistry B7.4

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level C2

Three students each prepare a standard solution by dissolving 10.6 g of solid from different bottles labelled 'sodium carbonate' in exactly 1 dm^3 of water. They use this standard solution in a titration to determine the exact concentration of a solution of sulfuric acid at approximately 0.1 mol dm^{-3} . They each use a pipette to measure out exactly 25.00 cm^3 of the standard solution into a conical flask, they each use the same indicator and they each carry out their titrations with great care and accuracy.

The volumes of sulfuric acid solution that they each use are listed below. Only student A finds the correct concentration of the sulfuric acid. Student B is within 20 % but student C is so far out that they know something is wrong. Student C asks for help and is reminded that some solids can contain water of crystallisation. Student A uses anhydrous sodium carbonate, but what is x in the formula $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}(\text{s})$ for students B and C?

Student A: 23.75 cm^3

Student B: 20.20 cm^3

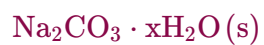
Student C: 8.80 cm^3

Part A

Acid concentration

Calculate the exact concentration of the sulfuric acid. Give your answer to 3 significant figures.

Part B



Find x in $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}(\text{s})$ for student B.

Find x in $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}(\text{s})$ for student C.

Question deck:

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LiH and PCl₃

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level C2

Reaction of 7.9 g of LiH with 45.8 g of PCl₃ produces 42.4 g of a white solid **A** and 8.0 dm³ of a colourless gas **B** (at RTP). Compound **B** (8.0 dm³) spontaneously ignites in air to give 32.7 g of a single compound **C**. A solution of **C** (9.8 g dissolved in 100 cm³) is neutralised by 150 cm³ of 2.0 mol dm⁻³ NaOH solution.

Part A

A

Deduce the formula of compound **A**.

Part B

B

Deduce the formula of compound **B**.

Part C

C

Deduce the formula of compound **C**.

Part D
Acid

Based on the number of acidic protons in **C**, what term would you use to describe this acid?

Adapted with permission from UCLES, STEP Chemistry, July 1999, Question 6

Question deck:

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Heating Calcium Hydroxide

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Heating calcium hydroxide

Write the equation for the action of heat on calcium hydroxide, including state symbols, balancing the equation with the lowest possible stoichiometric coefficients.

Part B

Decomposition of calcium hydroxide

Which of the following explains why magnesium hydroxide decomposes at a lower temperature than calcium hydroxide?

1. MgO has a larger magnitude lattice energy than CaO.
2. Mg has higher first and second ionisation energies than Ca.
3. $\text{Mg}(\text{OH})_2$ has a larger magnitude lattice energy than $\text{Ca}(\text{OH})_2$.

- ☐ 1, 2 and 3 are correct
- ☐ 1 and 2 only are correct
- ☐ 2 and 3 only are correct
- ☐ 1 only is correct
- ☐ 3 only is correct

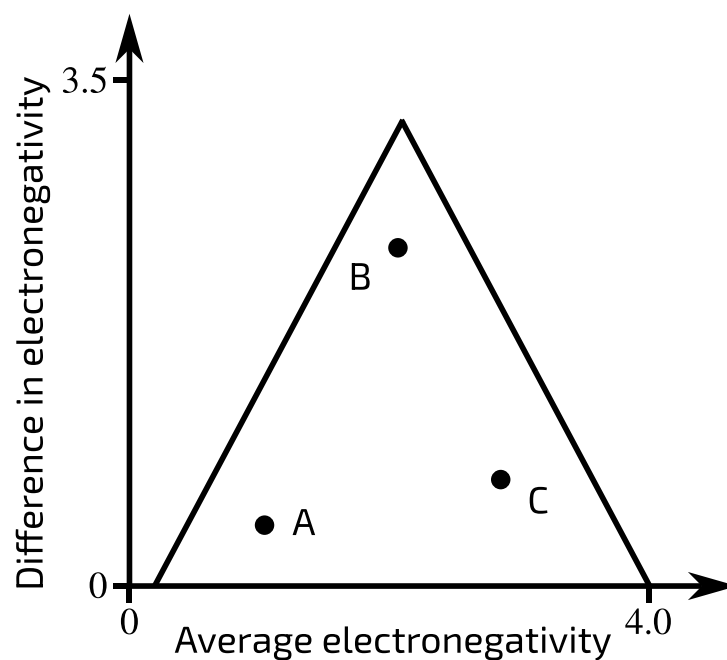
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Bonding Triangle

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C1

While we like to think of the various bonding types in chemistry as very different from one another, in reality metallic, covalent and ionic bonding exist on a continuum. One way of displaying this information is on the triangle below. Here, the x-axis scale spans the range of electronegativity values and we plot the average electronegativity of the elements involved in the compound, while we plot the difference of electronegativities on the y-axis.



Part A

Bonding types

In this triangle, metallic bonding is seen in the corner, covalent bonding in the corner and ionic bonding in the corner. Substances with predominantly metallic bonding have electronegativities of the elements involved. Covalent substances have differences in electronegativities between the elements involved. Ionic substances have differences in electronegativities between the elements involved.

Items:

Part B

Labelled positions

Identify the most likely positions in the triangle of the following binary compounds and hence deduce the predominant bonding type.

Binary compound	Label on diagram	Predominant bonding type
MgO		
CO ₂		
Al ₃ Mg ₂		

Items:

Part C
Corners

What diatomic gas will be found in the very bottom right corner of the triangle?

What element will be found in the very bottom left corner of the triangle?

What binary compound will you find in the very top corner of the triangle?

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Question deck:

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Platinum, Hydroquinone and Silver

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Part A

Oxidation numbers of platinum

The anti-cancer drug *cis-platin* has the formula $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$.

In the human body, one of the chloride ions of *cis-platin* is replaced by one water molecule.



What is the oxidation number of platinum in each of these complexes?

	<i>cis-platin</i>	<i>in the aquo complex</i>
A	+2	+1
B	+2	+2
C	+4	+3
D	+4	+4

- ☐ A
☐ B
☐ C
☐ D

Part B

Hydroquinone and silver bromide

When exposed film from a camera is developed, one step involves reacting the light-activated silver bromide crystals with hydroquinone in alkali according to the following equation:

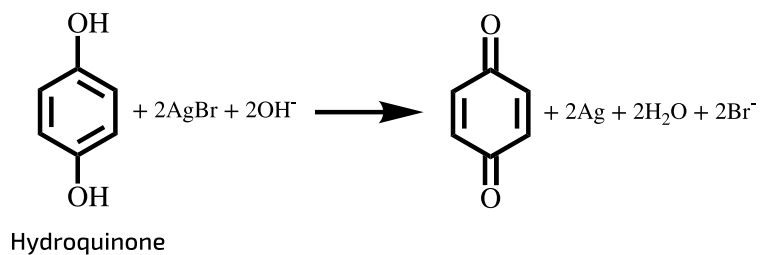


Figure 1: Hydroquinone and silver bromide in presence of base.

Which of the following describes the role of hydroquinone?

- ☐ It acts as both a base and a reducing agent
- ☐ It acts only as an acid
- ☐ It acts only as an oxidising agent
- ☐ It acts only as a reducing agent
- ☐ It acts as both an acid and a reducing agent

Part A adapted with permission from UCLES, A-Level Chemistry, June 1996, Paper 3, Question 9;

Part B adapted with permission from OCSEB, A-Level Chemistry, June 1994, Paper 1, Question 3

Question deck:

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Ferrite

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

Aerials in portable radios are made of a mixed oxide of calcium and iron known as 'Ferrite'. It contains 18.5 % calcium and 51.9 % iron by mass. Calculate the empirical formula of 'Ferrite' and hence deduce the oxidation number of the iron it contains.

Part A

Empirical formula

Empirical Formula:

Part B

Oxidation number

Oxidation number:

Adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 2, Question 3

Question deck:

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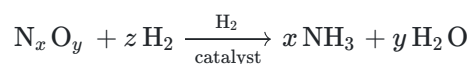
Oxides of Nitrogen

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C2

Part A

Oxides of nitrogen

In an attempt to establish the formula of an oxide of nitrogen, a known volume of the pure gas was mixed with hydrogen and passed over a catalyst at a suitable temperature. 100 % conversion of the oxide to ammonia and water was shown to have taken place.



2400 cm³ of the nitrogen oxide, measured at room temperature and pressure (RTP) produced 7.200 g of water. The ammonia produced was neutralised by 200 cm³ of 1.0 mol dm⁻³ HCl.

[Molar volume of gas at RTP = 24000 cm³ mol⁻¹]

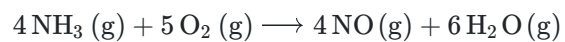
What was the oxidation number of nitrogen in the solid oxide?

- ☐ +1
- ☐ +2
- ☐ +3
- ☐ +4
- ☐ +5

Part B

Oxidation numbers of nitrogen

The key stage in the manufacture of nitric acid is the reaction of ammonia with air in the presence of a platinum-rhodium gauze:



What is the oxidation number of nitrogen in

NH₃

NO

Part A adapted with permission from UCLES, A-Level Chemistry, November 1989, Paper 3, Question 2;

Part B adapted with permission from UCLES, A-Level Chemistry, November 1995, Paper 3, Question 1

Question deck:

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Fractional Oxidation States

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

As oxidation states are a theoretical concept, they do not need to necessarily take integer values. In this question, we look at two examples.

Part A

Superoxide ion

The superoxide ion, O_2^- , consists of two identical oxygen atoms. What is the oxidation state of oxygen in this chemical species?

Part B

Magnetite

What is the average oxidation state of iron in magnetite, Fe_3O_4 ?

Part C

Meaning

Oxidation states can be thought of as the charges that we would see on the atoms if the bonding in a compound were purely ionic. This means that it is perhaps not that surprising to see a fractional oxidation state in a covalent species, such as superoxide. However, since the charges on individual ions need to be integers (as ions form by losing or gaining a whole number of electrons), how can we explain the presence of fractional oxidation states in an ionic species such as magnetite?

While the **average** oxidation state of iron in Fe_3O_4 is fractional, this is due to the presence of iron ions with different oxidation states in the ionic lattice. For every ions there is one ion.

Items:

two

three

four

Fe^{2+}

Fe^{3+}

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